Loss L has density

$$f(l) = \frac{ab}{a \cdot \exp(bl_0) + b \cdot \exp(al_0)} \cdot \left(\exp(1_{L \le l_0}) + \exp(1_{L \ge l_0})\right)$$

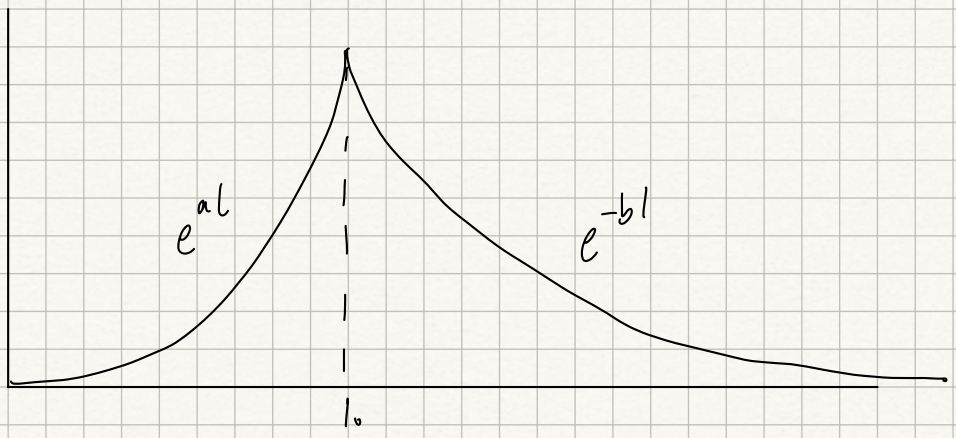
$$C = \frac{ab}{a \cdot \exp(bl_0) + b \cdot \exp(al_0)}, \quad \text{so that } f(l) \begin{cases} Ce^{al}, & L \le l_0 \\ Ce^{-bl}, & L > l_0 \end{cases}$$

$$\int_{-\infty}^{l_0} e^{al}\, dl = \left[\frac{e^{al}}{a}\right]_{-\infty}^{l_0} = \frac{e^{al_0}}{a}$$

$$\int_{l_0}^{\infty} e^{-bl}\, dl = \left[-\frac{e^{-bl}}{b}\right]_{l_0}^{\infty} = \frac{e^{-bl_0}}{b}$$

Therefore

$$\int f(l)\, dl = C\left(\frac{e^{al_0}}{a} + \frac{e^{-bl_0}}{b}\right) = C \cdot \frac{be^{al_0} + ae^{-bl_0}}{ab} = 1$$



CDF

for $l \le l_0$

$$F(l) = \int_{-\infty}^{l} Ce^{as}\, ds = \frac{C}{a} e^{al}$$

$$F(l_0) = \frac{C}{a} e^{al_0} = \frac{b^{al_0}}{ae^{-bl_0} + be^{al_0}} \quad / \div e^{al_0}$$

$$F(l_0) = \frac{b}{b + a e^{-(a+b)/l_0}}$$

Since F is non decreasing, $\alpha \geq F(l_0)$ implies α-quantile satisfies $VaR_\alpha(L) \geq l_0$

For $L > l_0$:

$$1 - F(L) = P(L > L) = \int_l^\infty C e^{-bs}\, ds =$$

$$= C\left[-\frac{e^{-bs}}{b}\right]_l^\infty = \frac{C}{b} e^{-bL}$$

Hence

$$F(L) = 1 - \frac{C}{b} e^{-bL}, \quad L > l_0$$

$$1 - \frac{C}{b} e^{-bx} = \alpha \iff \frac{C}{b} e^{-bx} = 1 - \alpha \iff e^{-bx} = \frac{b(1-\alpha)}{C}$$

Taking log and devide by $-b$

$$x = \frac{1}{b} \ln\left(\frac{C}{b(1-\alpha)}\right)$$

Subbing $\frac{C}{b} = \frac{a}{a e^{-bl_0} + b e^{al_0}}$

$$VaR_\alpha(L) = \frac{1}{b} \ln\left(\frac{\alpha}{(1-\alpha)(a e^{-bl_0} + b e^{al_0})}\right) =$$

$$= \frac{1}{b}\left[\ln\alpha - \ln(1-\alpha) - \ln(a e^{-bl_0} + b e^{al_0})\right] =$$

$$= -\frac{1}{b}\ln(1-\alpha) + \frac{1}{b}\ln\left(\frac{\alpha}{a e^{-bl_0} + b e^{al_0}}\right)$$

2.

$L \sim Bin(n,p)$, so L takes values in $\{0,1,\dots,n\}$

$$P(L=k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad F(k) = \sum_{j=0}^{k} \binom{n}{j} p^j (1-p)^{n-j}$$

F is right-continuous step functions. For generic α there is no x with $F(x) = \alpha$, so can't write $VaR_\alpha = F^{-1}(\alpha)$.

$$VaR_\alpha(L) = \inf\{x \in \mathbb{R} : F(x) \geq \alpha\}$$

Since F only increase at integers, the infinitum is attained at the integer

$$VaR_\alpha(L) = \min\left\{k \in \{0,1,\dots,n\} : \sum_{j=0}^{k} \binom{n}{j} p^j (1-p)^{n-j} \geq \alpha\right\}$$

Explicitly:

$n = 6$, $p = \frac{1}{2}$, $\alpha = 0.9$

With $p = \frac{1}{2}$, every outcome probability $\binom{6}{k}/2^6 = \binom{6}{k}/64$

$X = 64\,P(L=k)$ with $k = \{0,\dots,6\}$

$X = \{1, 6, 15, 20, 15, 6, 1\}$

$Y = 64\,F(k)$ with $k = \{0, \dots, 6\}$

$Y = \{1, 7, 22, 42, 57, 63, 64\}$

Smallest k with $F(k) \geq 0.9$, i.e. $64\,F(k) \geq 57.6$

$F(4) = \frac{57}{64} = 0.890625 < 0.9$ $\quad F(5) = \frac{63}{64} = 0.984375 \geq 0.9$

$$\text{VaR}_{0.9}(L) = 5.$$

Y and Z are indepind. with $E[Y] = 1/\lambda$ and $E[Z] = 1/\theta$, i.e.

$$f_Y(y) = \lambda e^{-\lambda y} 1_{y>0}, \quad f_Z(z) = \theta e^{-\theta z} 1_{z>0}$$

Let $L = \frac{Y}{Z}$ since Y and Z both >0, $\therefore L > 0$ so $F_L(l) = 0$ for $l \leq 0$ and we work on $l > 0$.

For $l > 0$, the event $\{Y/Z \leq l\}$ equivalent to $\{Y \leq lz\}$

$$F_L(l) = \mathbb{P}(Y \leq lZ) = \int_0^\infty \mathbb{P}(Y \leq lz) f_Z(z)\, dz = \int_0^\infty (1 - e^{-\lambda l z}) \theta e^{-\theta z}\, dz$$

since $\theta \int_0^\infty e^{-\theta z} dz = 1$

$$\theta \int_0^\infty e^{-(\theta + \lambda l)z}\, dz = \frac{\theta}{\theta + \lambda l}$$

which converge since $\theta + \lambda l > 0$, so $F(l) = 1 - \frac{\theta}{\theta + \lambda l} = \frac{\lambda l}{\theta + \lambda l}$, $l > 0$

$F_L(0^+) = 0$, $F_L(\infty) = 1$, so $VaR_\alpha(L) = F_L^{-1}(\alpha)$

$F_L(x) = \alpha$ for $\alpha \in (0,1)$

$$\frac{\lambda x}{\theta + \lambda x} = \alpha \iff \lambda x = \alpha\theta + \alpha\lambda x \iff \lambda x(1-\alpha) = \alpha\theta$$

$$VaR_\alpha(L) = \frac{\theta}{\lambda} \cdot \frac{\alpha}{1-\alpha} \quad , \alpha \in (0,1)$$